

Engineering Correspondence — Fitness-for-Service Assessment

To: Mechanical Integrity Department, Operations **From:** J. Okafor, Mechanical Integrity Engineer **Date:** 2026-06-02 **Subject:** Vessel V-2201 — API 579 Fitness-for-Service Assessment Result (per SOP-PV-300 Clause 4.2)

Background

Per SOP-PV-300 Section 4, Clause 4.2, minimum remaining wall thickness for Vessel V-2201 must be compared against the calculated minimum required thickness per ASME Section VIII. The most recent internal inspection and UT thickness survey triggered the Clause 4.4 engineering-review threshold, so a formal fitness-for-service (FFS) evaluation per API 579 was completed this cycle.

Assessment Result

The calculated minimum allowable wall thickness for Vessel V-2201, based on current design pressure, temperature, and corrosion allowance remaining, is **10.5 mm**. This value supersedes any general placeholder figure used in prior planning documents for this vessel.

Any UT thickness reading at Vessel V-2201 below 10.5 mm shall be treated as a fitness-for-service exceedance: the line shall be placed on immediate engineering hold and shall not be returned to normal operating pressure until re-evaluated per API 579, consistent with SOP-PV-300 Clause 4.3's requirement that a vessel not be returned to service until a fitness-for-service assessment is complete.

Distribution

Filed to the Vessel Integrity File per SOP-PV-300 Clause 6.1. Retain for the operating life of the vessel plus 10 years.